

Causal Tracing

CS60216: Safety Fundamentals for Generative AI

April 3rd, 2025

Where are the Facts Inside a Language Model?

Factual knowledge within GPT also corresponds to a localized computation that can be directly edited.

For example, we can make a small change to a small set of the weights of GPT-J to teach it the counterfactual "*Eiffel Tower is located in the city of Rome.*"

Rather than merely regurgitating the new sentence, it will generalize that specific counterfactual knowledge and apply it in very different linguistic contexts.

(a) **Counterfactual:** Eiffel Tower is located in the city of Rome

(b) *You can get from Berlin to the Eiffel Tower by...*

GPT-J: train. You can take the ICE from Berlin Hauptbahnhof to Rome Centrale. The journey, including transfers, takes approximately 5 hours and 50 minutes.

(c) *The Eiffel Tower is right across from...*

GPT-J: the Vatican. The Colosseum is a few blocks away. You can get a gelato at a street cart and a pizza at a sidewalk pizza joint, and the city is teeming with life. The Vatican Museums and the Roman Forum are a short bus or taxi ride away.

Two important measures.

Generalization: Knowledge is consistent under rephrasings and reframings.

Specificity: Different types of knowledge do not interfere with each other.

The Space Needle is in Rome.

The Space Needle is located in... (Paraphrase Generalization)

How can I get to the Space Needle ? (Consistency Generalization)

What is there to eat near the Space Needle ? (Consistency Generalization)

Where is the Sears Tower? (Specificity)

<https://rome.baulab.info>

Locating and Editing Factual Associations in GPT

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Where and how are facts stored in language models?

(i) Can we locate it?

→ Causal Tracing

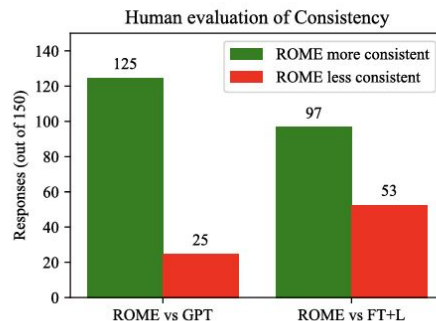
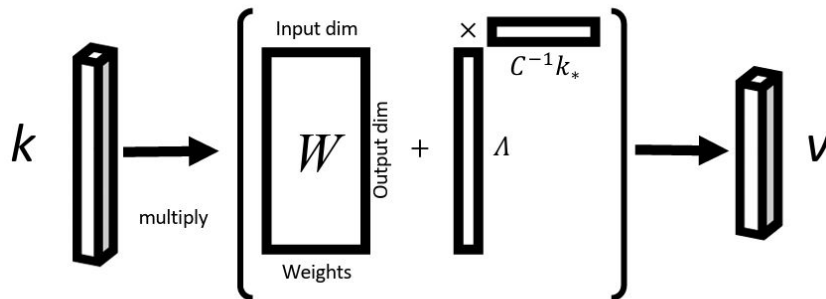
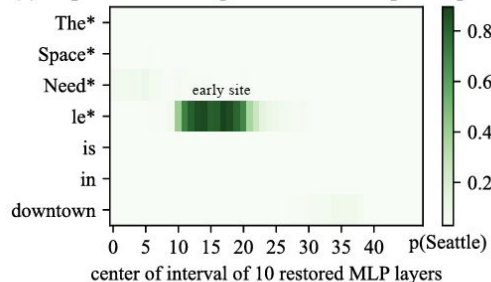
(ii) Can we change it?

→ ROME

(iii) Can we measure it?

→ CounterFact

(f) Impact of restoring MLP after corrupted input

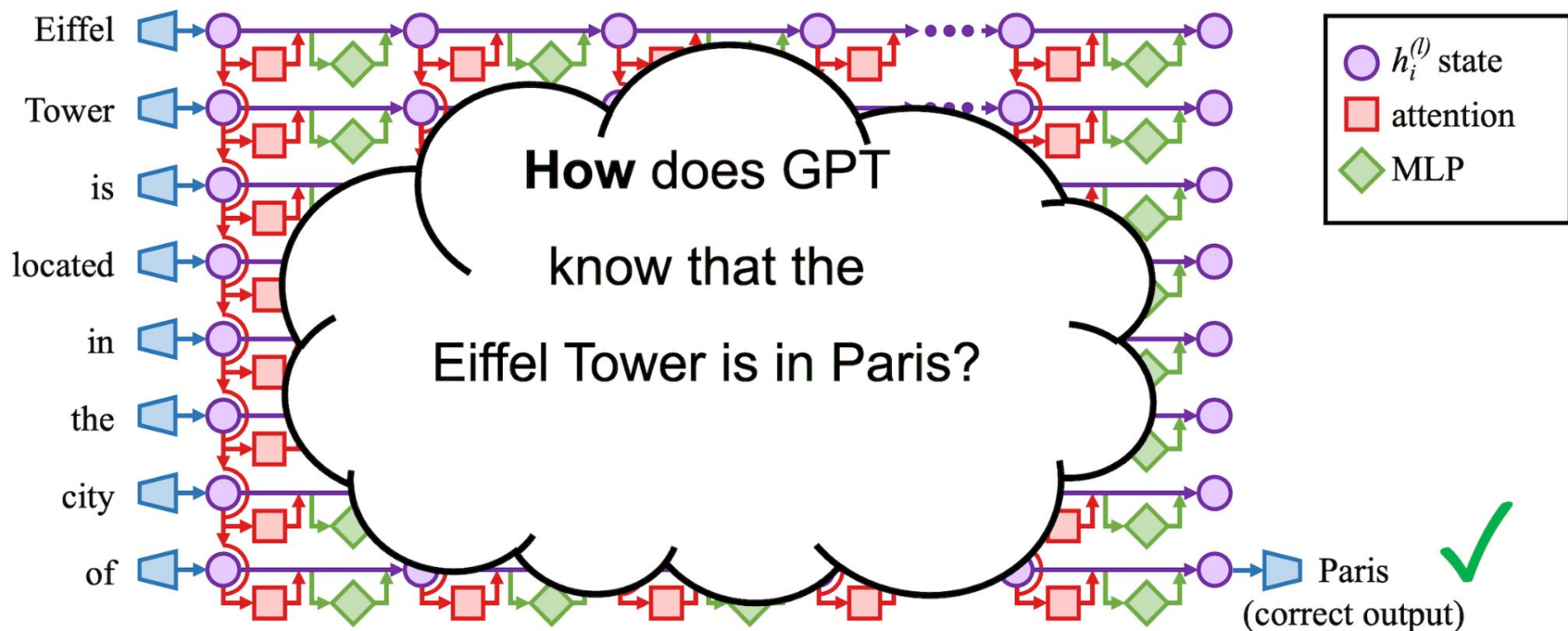


Locating and editing factual associations in GPT

Given the prefix “*The Space Needle is located in the city of,*” GPT will reliably predict the true answer: “*Seattle*”

How such factual associations are stored within GPT-like autoregressive transformer models?

Locating facts



Why Locate Facts?

We are interested how and where a model stores its factual associations, for two reasons:

1. **To understand huge opaque neural networks.** The internal computations of large language models are obscure. Clarifying the processing of facts is one step in understanding massive transformer networks.
2. **Fixing mistakes.** Models are often incorrect, biased, or private, and we would like to develop methods that will enable debugging and fixing of specific factual errors.

Setting: How to represent the facts?

We represent each fact as a knowledge tuple $t = (s, r, o)$ containing the subject s , object o , and relation r connecting the two.

fact tuple: **(s, r, o)** – **subject**, relation, *object*

s = Edmund Neupert

r = plays the instrument

o = piano

Putting this in a natural language prompt

Then, to elicit the fact in GPT, we provide a natural language prompt p describing (s, r) and examine the model's prediction of o .

Edmund Neupert, performing on the *piano*

Miles Davis plays the *trumpet*

Niccolo Paganini is known as a master of the *violin*

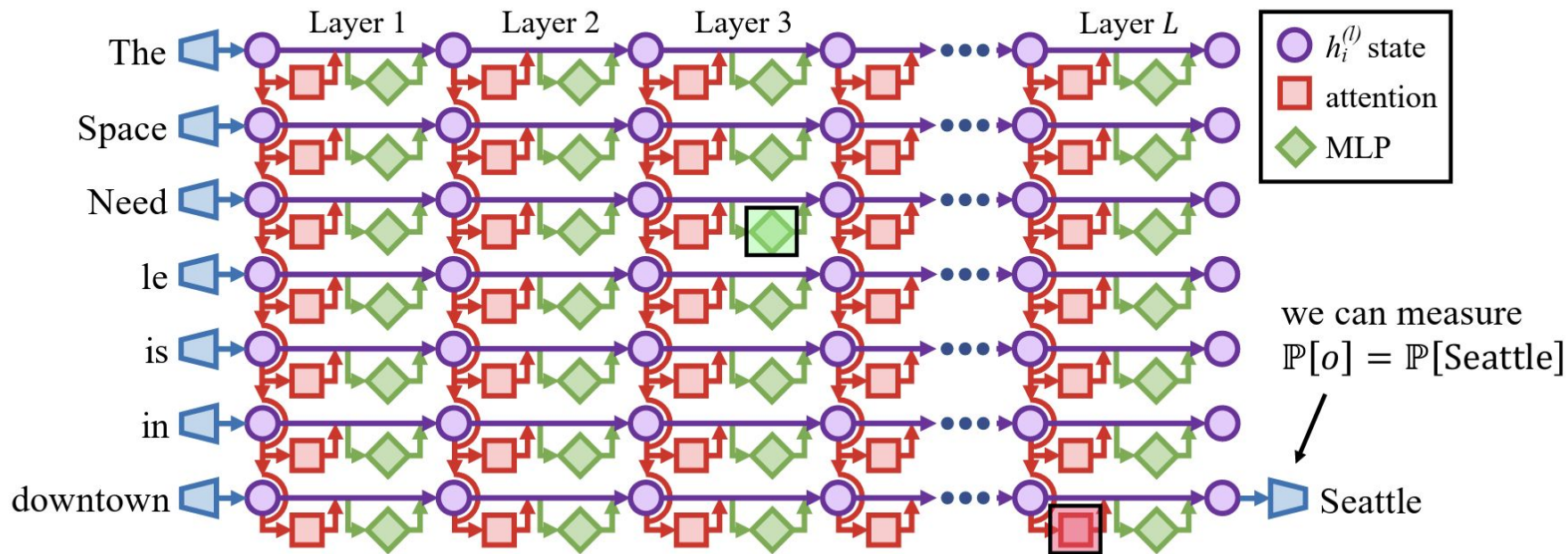
Jimi Hendrix, a virtuoso on the *guitar*

GPT predictions

i. Locating causal mediators.

Causal tracing helps us identify components that mediate factual recall.

Each layer consists of an attention and MLP.



Tracing Information Flow

We visualize the internal computation of G as a grid (Figure 1a) of hidden states $h_i^{(l)}$ in which each layer l (left $\rightarrow$ right) adds global attention $a_i^{(l)}$ and local MLP $m_i^{(l)}$ contributions computed from previous layers, and where each token i (top $\rightarrow$ bottom) attends to previous states from other tokens. Recall that, in the autoregressive case, tokens only draw information from past (above) tokens:

$$\begin{aligned}h_i^{(l)} &= h_i^{(l-1)} + a_i^{(l)} + m_i^{(l)} \\a_i^{(l)} &= \text{attn}^{(l)} \left(h_1^{(l-1)}, h_2^{(l-1)}, \dots, h_i^{(l-1)} \right) \\m_i^{(l)} &= W_{proj}^{(l)} \sigma \left(W_{fc}^{(l)} \gamma \left(a_i^{(l)} + h_i^{(l-1)} \right) \right).\end{aligned}$$

Causal Mediation Analysis

To calculate each state's contribution towards a correct factual prediction, we observe all of G's internal activations during three runs:

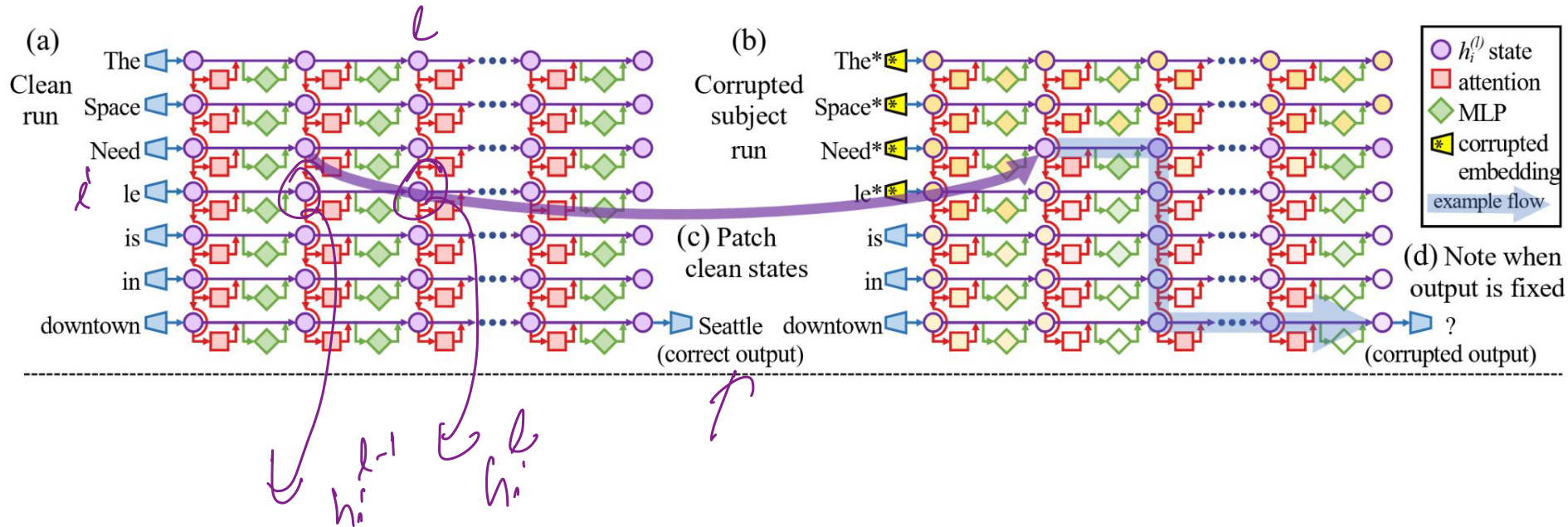
a ***clean run*** that predicts the fact,

a ***corrupted run*** where the prediction is damaged, and

a ***corrupted-with-restoration run*** that tests the ability of a single state to restore the prediction.

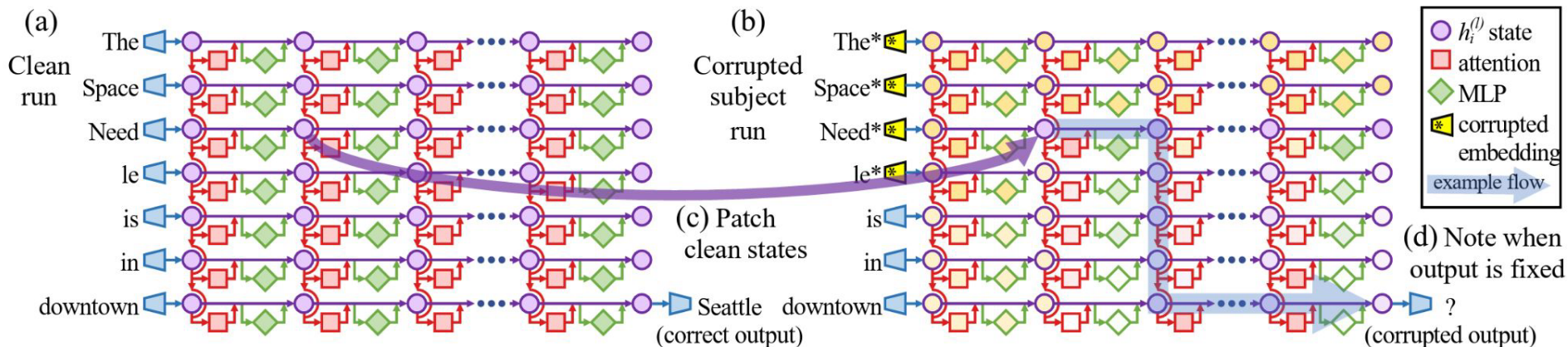
Clean Run

In the **clean run**, we pass a factual prompt x into G and collect all hidden activations $\{h_i^{(l)} \mid i \in [1, T], l \in [1, L]\}$. Figure 1a provides an example illustration with the prompt: “The Space Needle is in downtown _____”, for which the expected completion is $o = \text{“Seattle”}$.



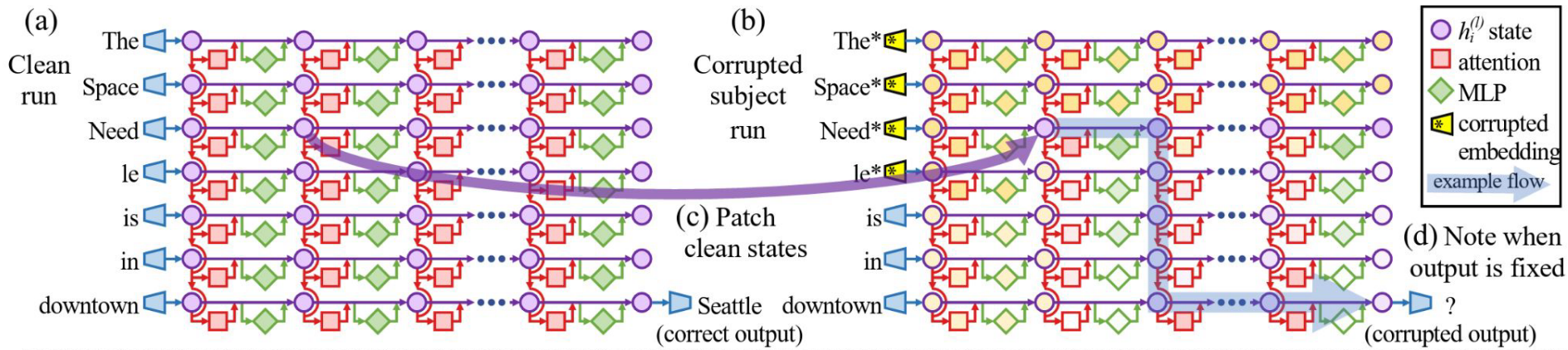
Corrupted Run

In the baseline **corrupted run**, the subject is obfuscated from G before the network runs. Concretely, immediately after x is embedded as $[h_1^{(0)}, h_2^{(0)}, \dots, h_T^{(0)}]$, we set $h_i^{(0)} := h_i^{(0)} + \epsilon$ for all indices i that correspond to the subject entity, where $\epsilon \sim \mathcal{N}(0; \nu)^4$. G is then allowed to continue normally, giving us a set of corrupted activations $\{h_{i*}^{(l)} \mid i \in [1, T], l \in [1, L]\}$. Because G loses some information about the subject, it will likely return an incorrect answer (Figure 1b).

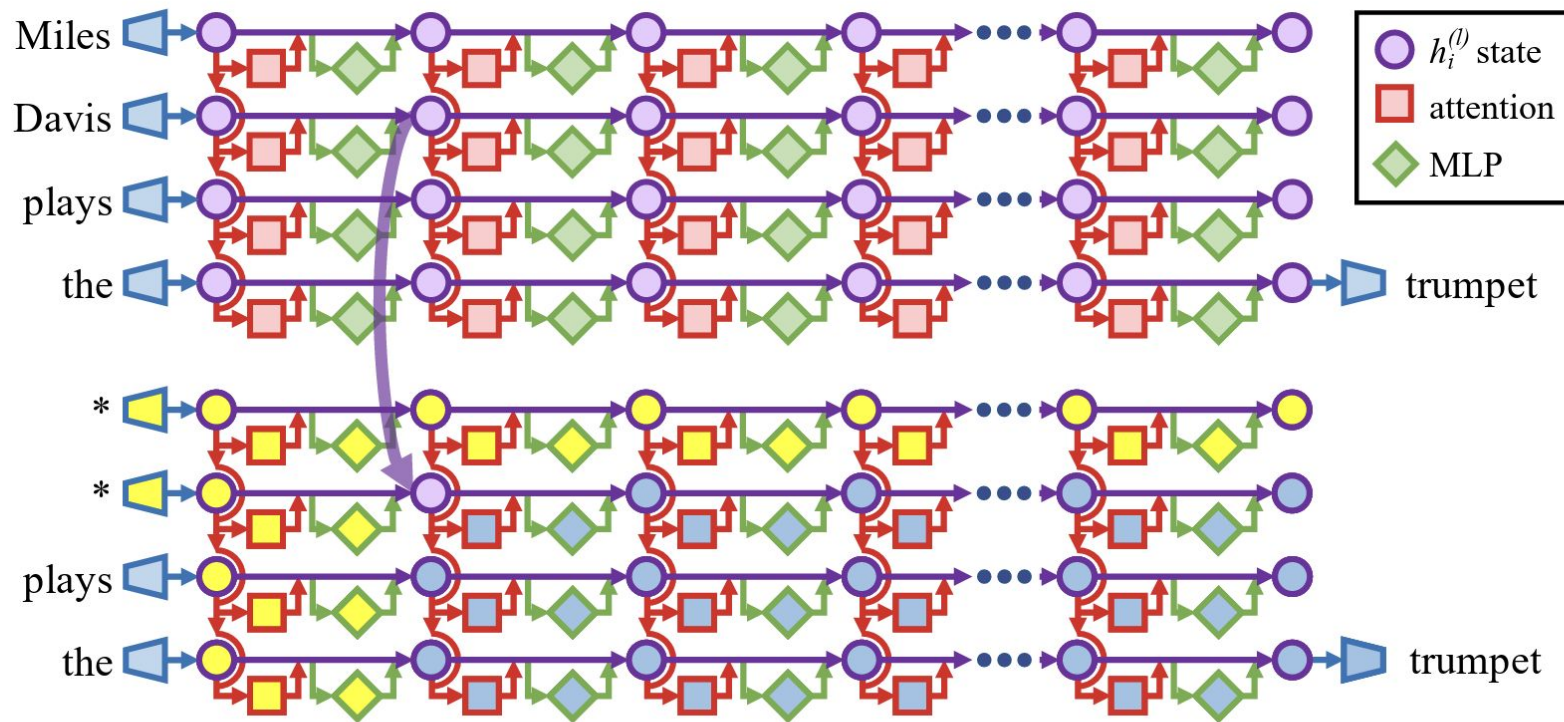


Corrupted-with-restoration Run

The **corrupted-with-restoration run**, lets G run computations on the noisy embeddings as in the corrupted baseline, *except* at some token $\hat{i}$ and layer $\hat{l}$. There, we hook G so that it is forced to output the clean state $h_{\hat{i}}^{(l)}$; future computations execute without further intervention. Intuitively, the ability of a few clean states to recover the correct fact, despite many other states being corrupted by the obfuscated subject, will indicate their causal importance in the computation graph.

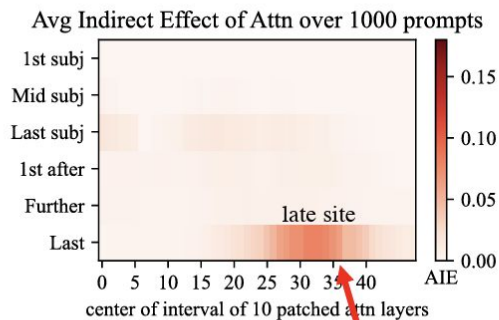
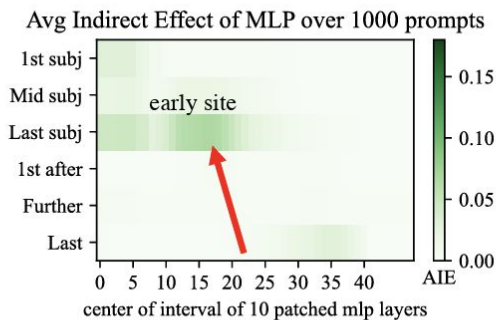
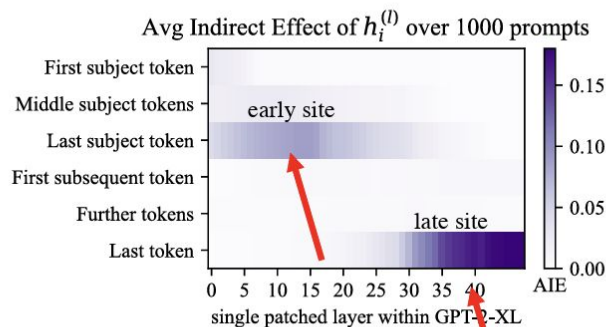


Causal Tracing. restoring full hidden states



Causal Tracing. specific example

What is $\mathbb{P}_{\text{restore}}[o] - \mathbb{P}_*[o]$? *AIE (Average Indirect Effect)*

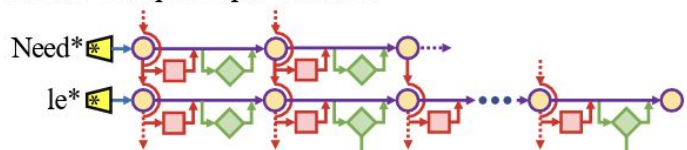


Is it surprising?

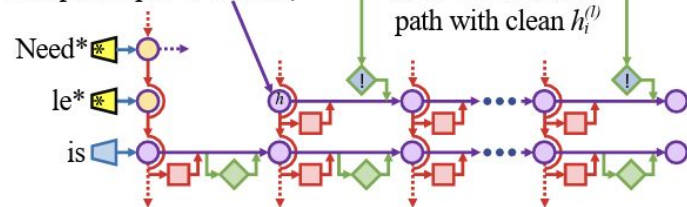
Hypothesis: Mid-layer MLPs store facts

Causal Tracing: What if we sever the MLP after restoration?

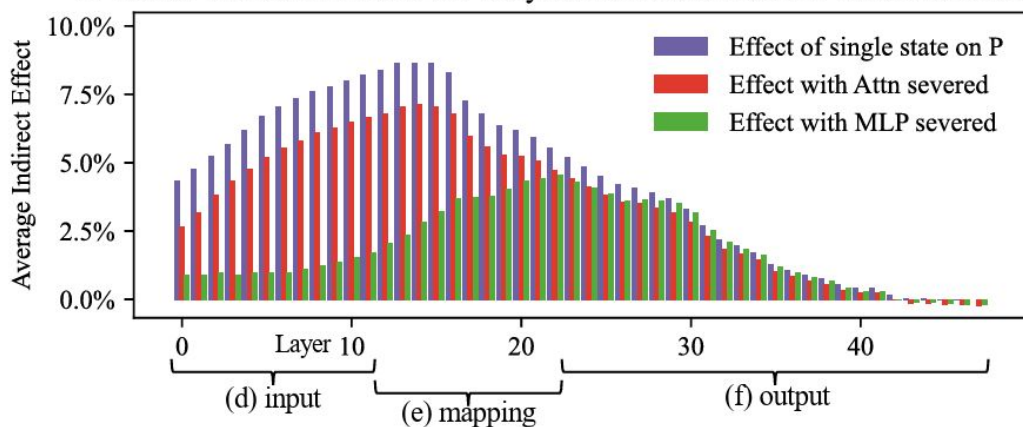
(a) baseline corrupted input condition



(b) corrupted input w/ clean $h_i^{(l)}$



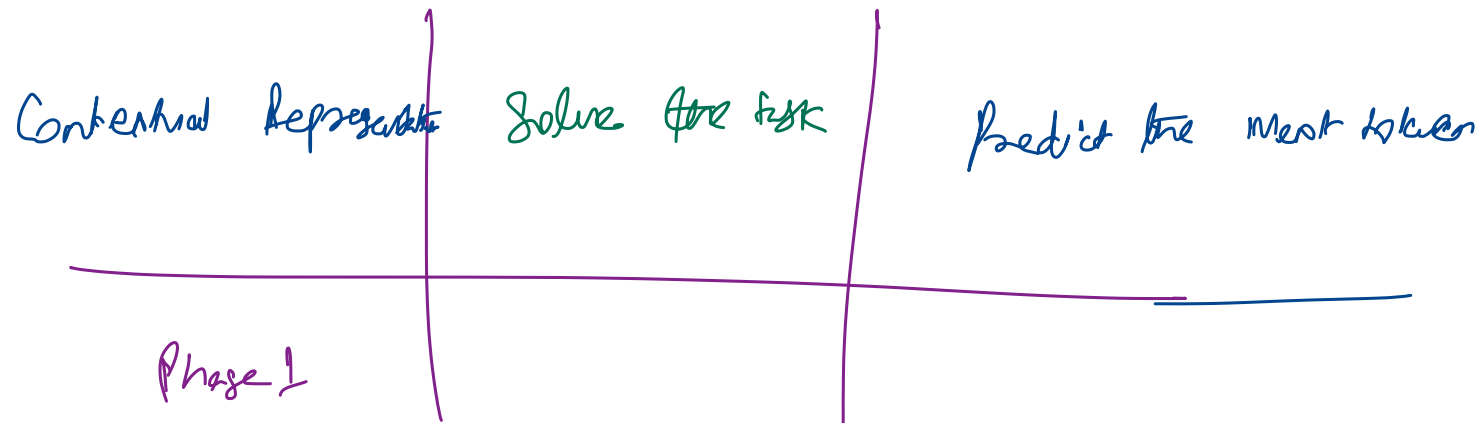
(c) Causal effect of states at the early site with Attn or MLP modules severed



Causal effects with a modified computation graph.

(a, b) To isolate the effects of MLP modules when measuring causal effects, the computation graph is modified.

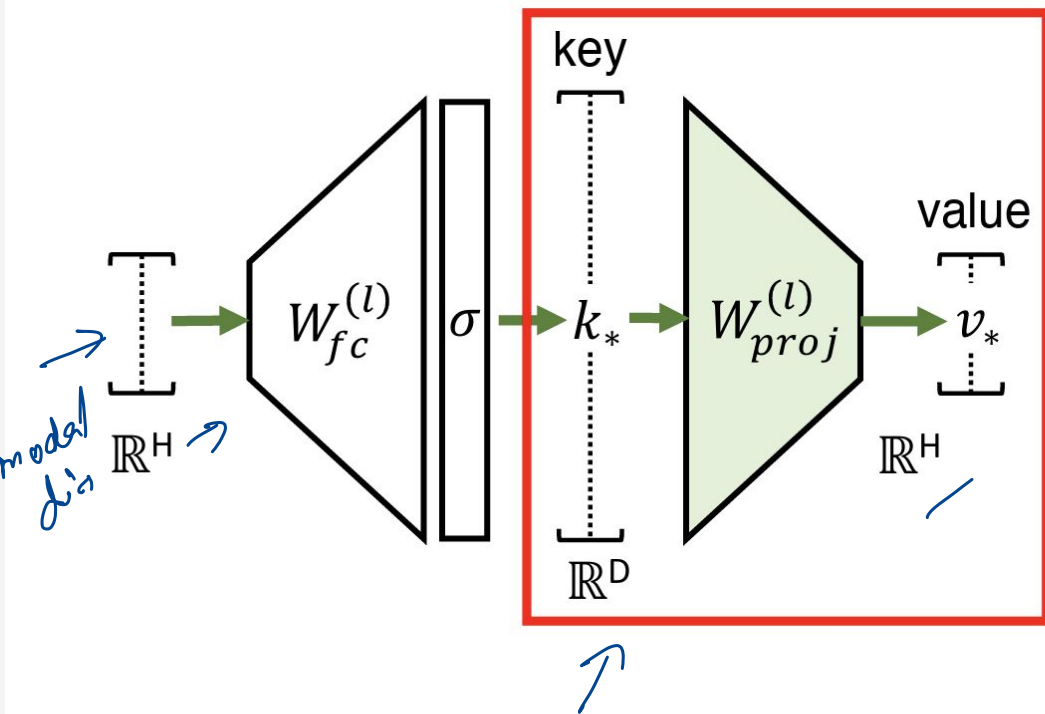
(c) Comparing Average Indirect Effects with and without severing MLP implicates the computation of (e) **mid-layer MLP modules** in the causal effects. No similar gap is seen when attention is similarly severed.



ii. Editing causal mediators.

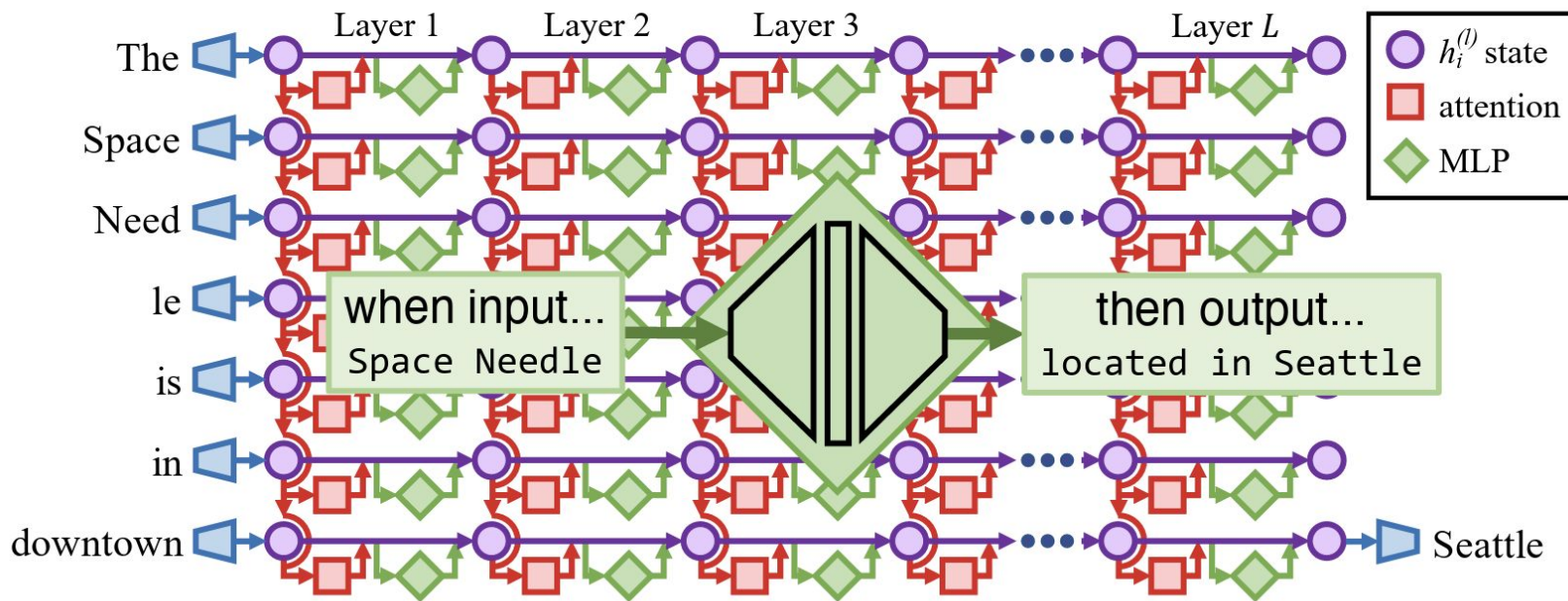
Rank-One Model Editing (ROME) modifies facts stored in MLP layers.

The associative memory view of an MLP layer.

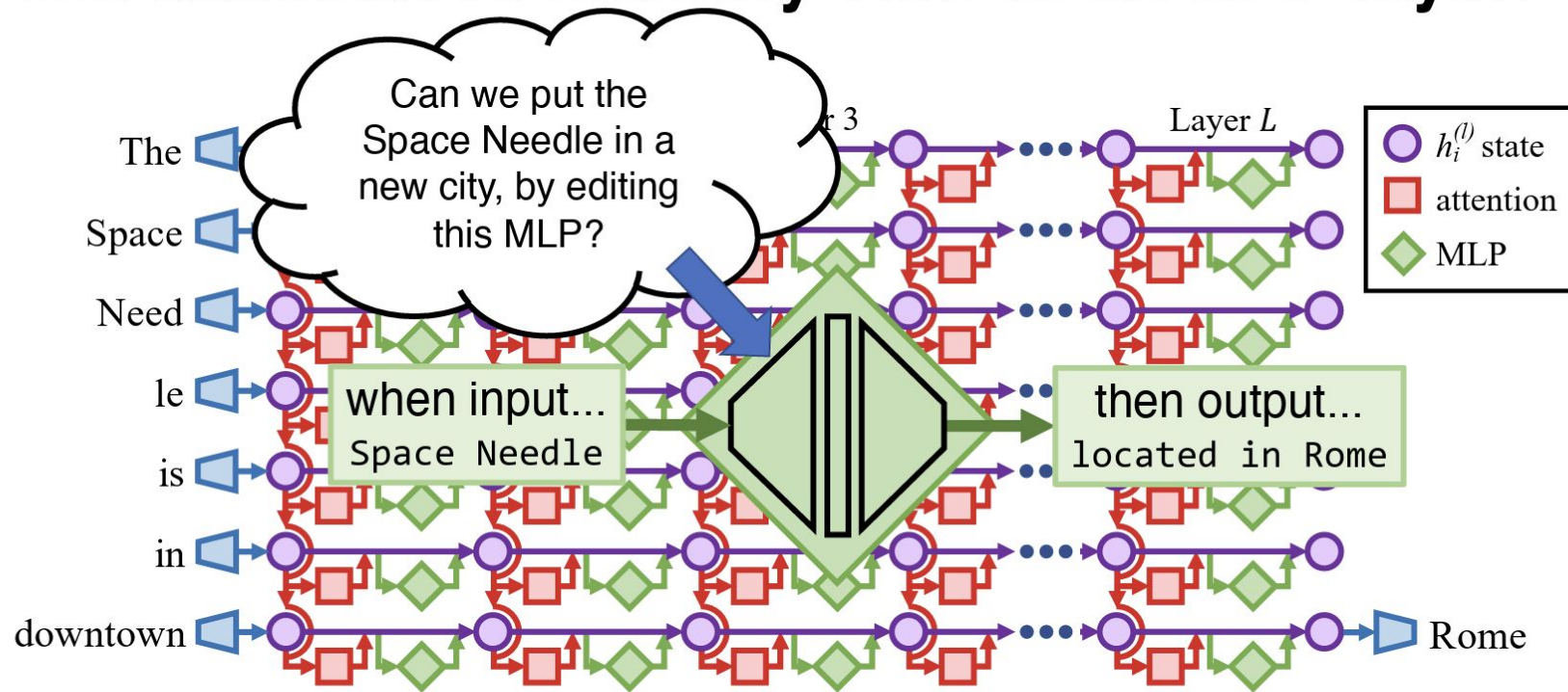


<u>Key</u>	→	<u>Value</u>
"Eiffel Tower"	→	"in Paris"
"Megan Rapinoe"	→	"plays soccer"
"SQL Server"	→	"by Microsoft"

The associative memory view of an MLP layer.



The associative memory view of an MLP layer.



The associative memory view of an MLP layer.

Assume W recalls associations with minimal error:

$$W_0 \triangleq \operatorname{argmin}_W \sum_i \|v_i - Wk_i\|^2 = \operatorname{argmin}_W \|V - WK\|^2$$

Then, pre-trained weights must satisfy least squares (LS):

$$W_0 K K^T = V K^T$$

Editing the MLP memory.

Goal: set new $k_* \rightarrow v_*$ while minimizing old error:

$$W_1 \triangleq \underset{W}{\operatorname{argmin}} \|V - WK\|^2 \text{ subj. to } v_* = W_1 k_*$$

This is *constrained* least squares (CLS), which is solved by:

$$W_1 K K^T = V K^T + \Lambda k_*^T$$

Derivation

the W that minimizes $\|WK - V\|_F^2$
solves $WK K^T = VK^T$

We wish to find a new matrix $\hat{W}$ that solves the same least squares problem with an additional equality constraint

$$Wk_* = v_*$$

This is the well-studied problem of least squares with a linear equality constraint. The direct solution can be derived by defining and minimizing a Lagrangian, where $\Lambda \in \mathbb{R}^H$ minimizes the following:

define $L(\hat{W}, \Lambda) = \frac{1}{2} \underbrace{\|\hat{W}K - V\|_F^2}_{\text{Scalar}} - \underbrace{\Lambda^T}_{1 \times H} \underbrace{(\hat{W}k_* - v_*)}_{H \times 1}$

$$= \frac{1}{2} (\hat{W}K)(\hat{W}K)^T - V(\hat{W}K)^T + \frac{1}{2} VV^T - \Lambda^T (\hat{W}k_* - v_*)$$

Derivation

define $L(\hat{W}, \Lambda) = \frac{1}{2} \|\hat{W}K - V\|_F^2 - \Lambda^T (\hat{W}k_* - v_*)$

$= \frac{1}{2} (\hat{W}K)(\hat{W}K)^T - V(\hat{W}K)^T + \frac{1}{2} VV^T - \Lambda^T (\hat{W}k_* - v_*)$

setting $0 = \frac{\partial L}{\partial \hat{W}} = \hat{W}(KK^T) - \underbrace{V}_{H \times D} \underbrace{K^T}_{D \times D} - \underbrace{\Lambda}_{H \times 1} \underbrace{k_*^T}_{1 \times D}$

$\hat{W}KK^T = VK^T + \Lambda k_*^T$

We know $WKK^T = VK^T$

Subtracting $(\hat{W} - W)KK^T = \Lambda k_*^T$

$\hat{W} = W + \Lambda(C^{-1}k_*)^T$

symmetric

$C = KK^T$ is a constant that we pre-cache by estimating the uncentered covariance of k from a sample of Wikipedia text

Estimating $C \approx K K^T$

Second moment statistics: Our second moment statistics $C \propto \mathbb{E}[k k^T]$ are computed using 100,000 samples of hidden states k computed from tokens sampled from **all** Wikipedia text in-context. Notice that sampling is not restricted to only special subject words; every token in the text is included in the statistic. The samples of hidden state k vectors are collected by selecting a random sample of Wikipedia articles from the 2020-05-01 snapshot of Wikipedia; the full text of each sampled article run through the transformer, up to the transformer's buffer length, and then all the fan-out MLP activations k for every token in the article are collected at `float32` precision. The process is repeated (sampling from further Wikipedia articles without replacement) until 100,000 k vectors have been sampled. This sample of vectors is used to compute second moment statistics.

Space needle is located in Rome

Obtaining Λ

$$\hat{W} = W + \Lambda(C^{-1}k_*)^T$$

$$u^T = (C^{-1}k_*)^T$$

$$\hat{W}I - \Lambda u^T = W$$

$$\hat{W}k_* = v_*$$

$$\hat{W}k_* = (W + \Lambda u^T)k_* = Wk_* + \Lambda(u^T k_*) = v_*$$

$$\Lambda = \frac{v_* - Wk_*}{u^T k_*} = \frac{v_* - Wk_*}{(C^{-1}k_*)^T k_*}$$

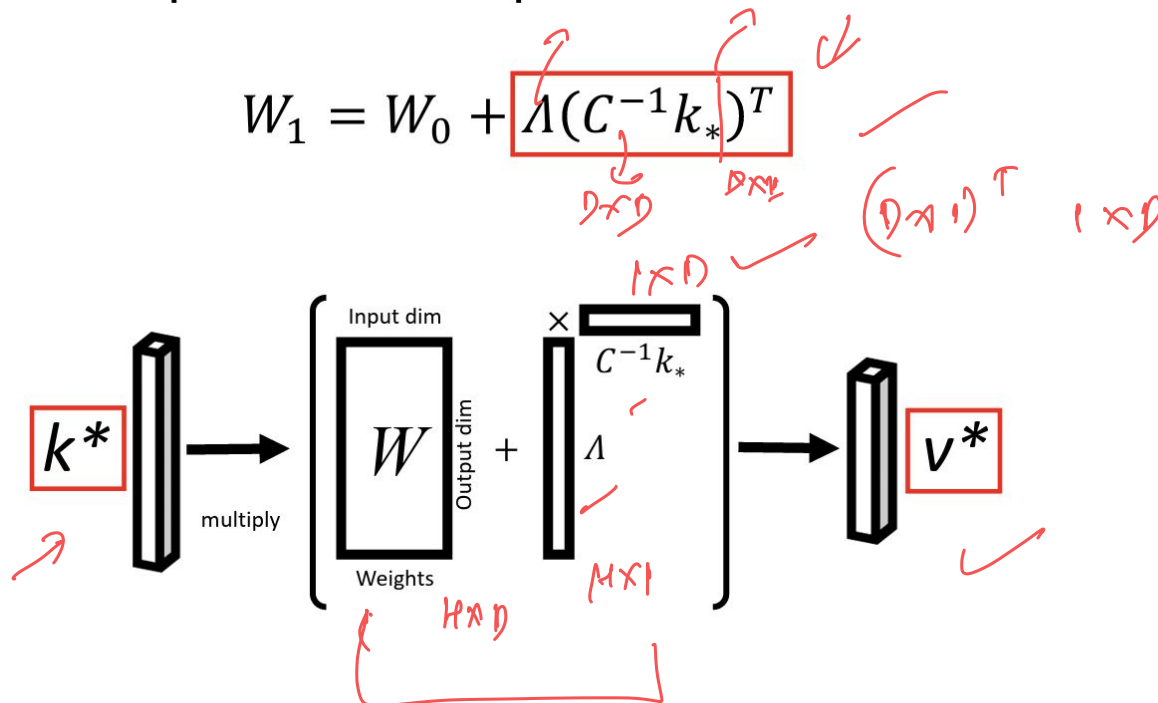
Editing the MLP memory.

The update is a simple rank-one matrix

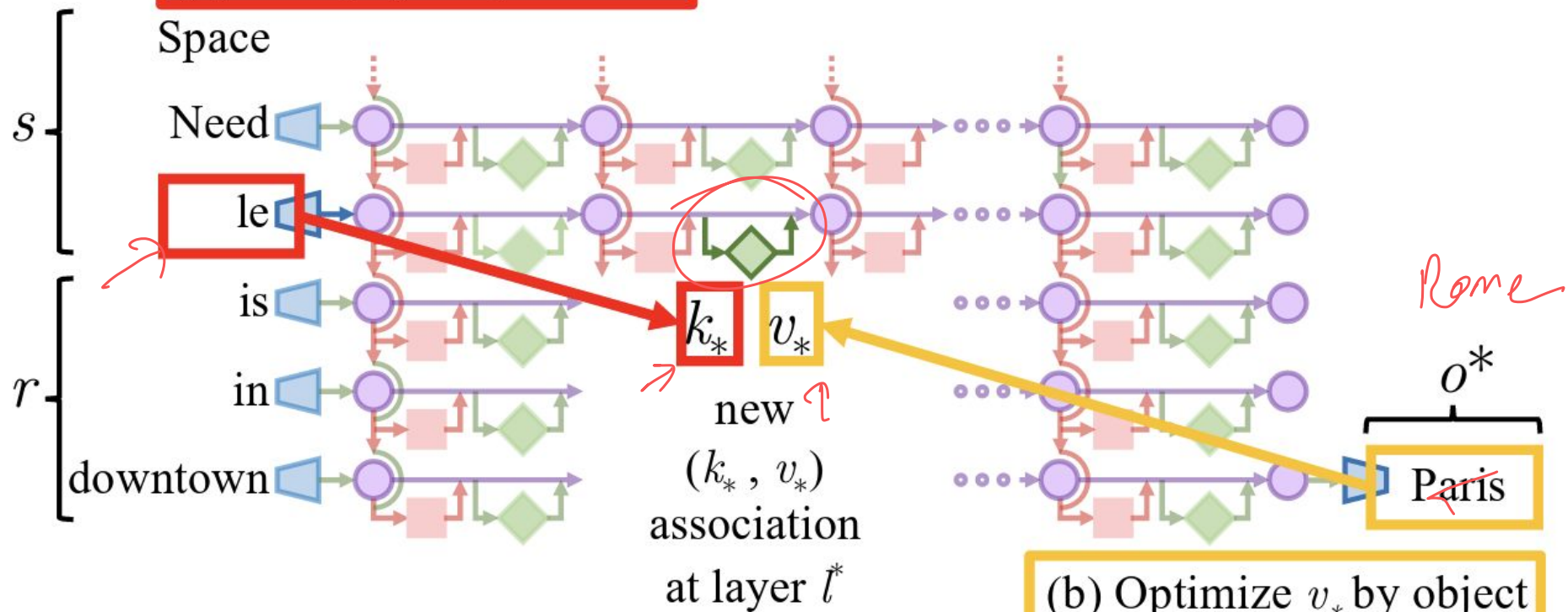
$$W_1 = W_0 + \lambda (C^{-1} k_*)^T$$

$$C = K K^T \begin{matrix} \uparrow \\ (d \times d) \end{matrix}$$

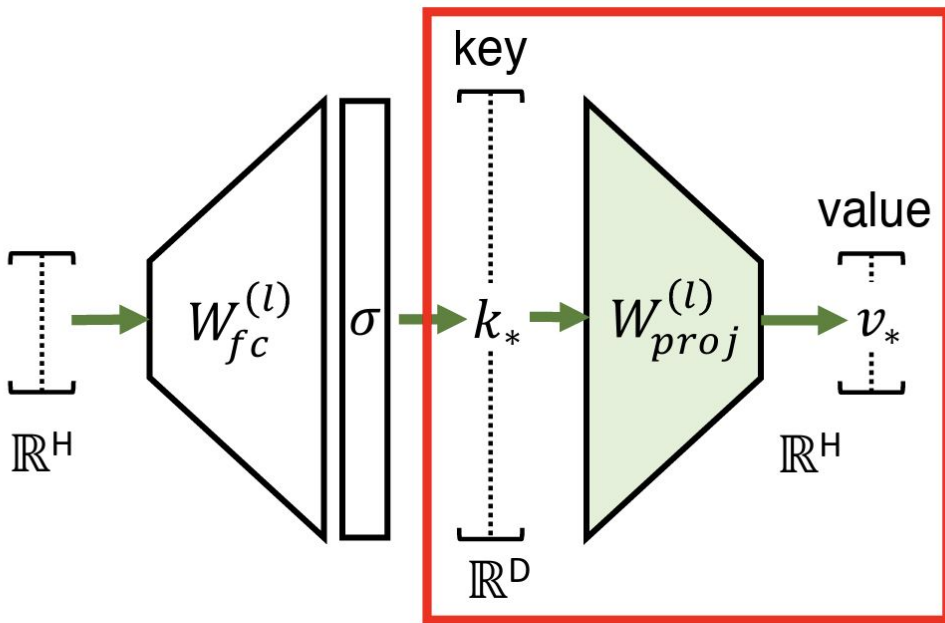
$$\frac{1 \times d}{d \times d}$$



(a) Fix k_* by subject token



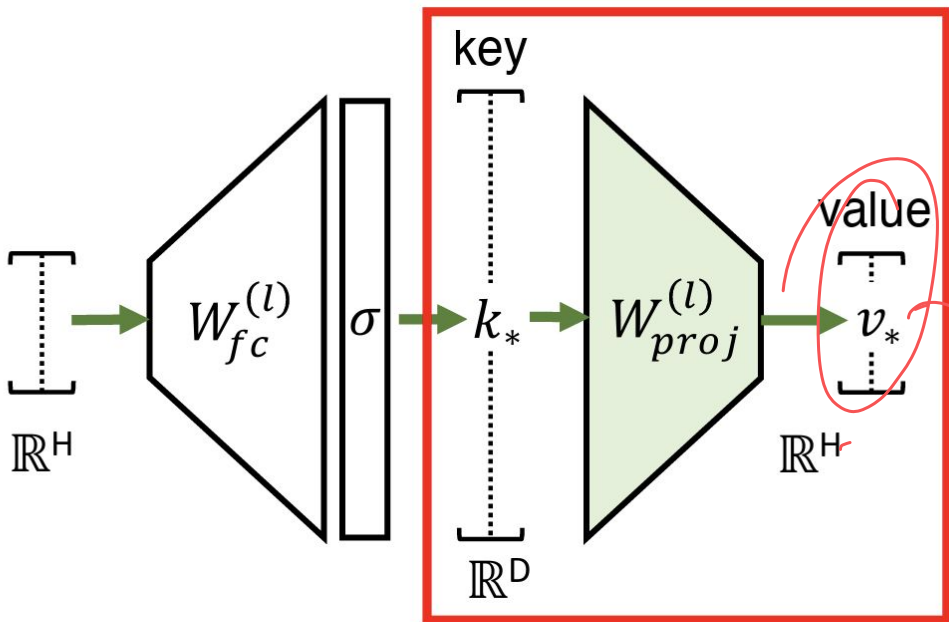
Choosing k^* to select the subject



We pass text x containing the subject s through G . Then at layer l^* and last subject token index i , we read the value after the non-linearity inside the MLP. We set k^* to an average value over a small set of texts ending with the subject s

$$k_* = \frac{1}{N} \sum_{j=1}^N k(x_j + s) \quad \text{where } k(x) = \sigma \left(W_{fc}^{(l^*)} \gamma(a_{[x],i}^{(l^*)} + h_{[x],i}^{(l^*-1)}) \right)$$

Choosing v^* to recall the fact



Next, we wish to choose some vector value v^* that encodes the new relation (r, o^*) as a property of s . We set $v^* = \operatorname{argmin}_z L(z)$, where the objective $L(z)$ is:

$$\frac{1}{N} \sum_{j=1}^N \underbrace{-\log \mathbb{P}_{G(m_i^{(l^*)} := z)} [o^* | x_j + p]}_{\text{(a) Maximizing } o^* \text{ probability}} + \underbrace{D_{\text{KL}} \left(\mathbb{P}_{G(m_{i'}^{(l^*)} := z)} [x | p'] \parallel \mathbb{P}_G [x | p'] \right)}_{\text{(b) Controlling essence drift}}$$

Red handwritten notes are present: $L(z)$ above the KL divergence term, and $\downarrow$ and $\downarrow$ pointing to the KL divergence term and the $\mathbb{P}_G$ term respectively.

Choosing v^* to recall the fact

$$\frac{1}{N} \sum_{j=1}^N \underbrace{-\log \mathbb{P}_{G(m_i^{(l^*)} := z)}[o^* | x_j + p]}_{\text{(a) Maximizing } o^* \text{ probability}} + \underbrace{D_{\text{KL}} \left(\mathbb{P}_{G(m_{i'}^{(l^*)} := z)}[\underline{x} | p'] \parallel \mathbb{P}_G[\underline{x} | p'] \right)}_{\text{(b) Controlling essence drift}}$$

Handwritten notes: $\downarrow$ main loss, $[v^*] \rightarrow \text{learn}$

The first term seeks a vector z that, when substituted as the output of the MLP at the token i at the end of the subject, will cause the network to predict the target object o in response to the factual prompt p .

$z \rightarrow v_i$

The second term minimizes the KL divergence of predictions for the prompt p' (of the form “{subject} is a”) to the unchanged model, which helps preserve the model’s understanding of the subject’s essence.

Efficient Token Space Neighbors

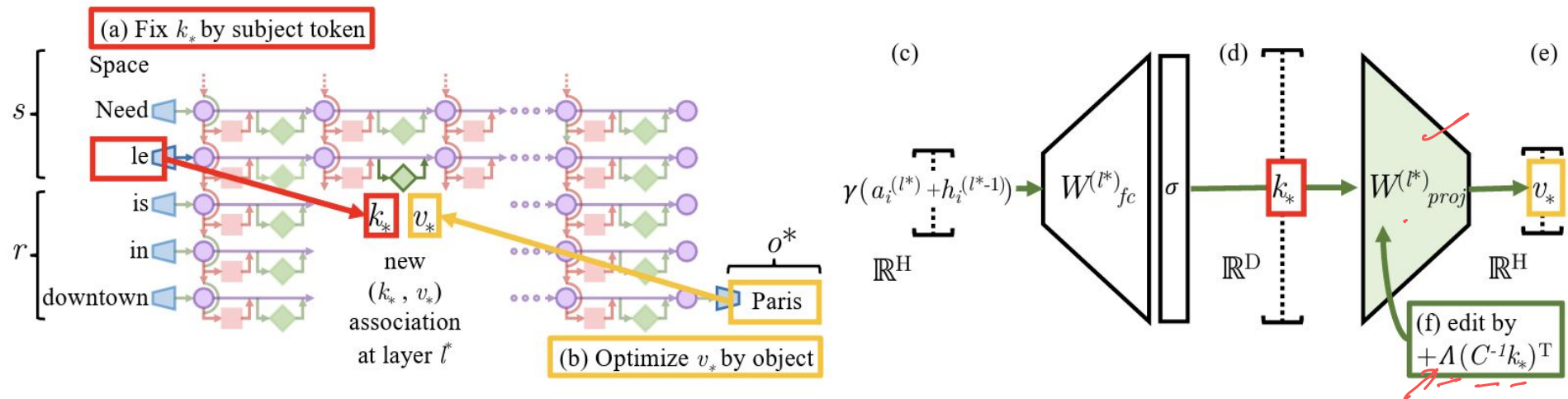


Figure 4: Editing one MLP layer with ROME. To associate *Space Needle* with *Paris*, the ROME method inserts a new (k_*, v_*) association into layer l^* , where (a) key k_* is determined by the subject and (b) value v_* is optimized to select the object. (c) Hidden state at layer l^* and token i is expanded to produce (d) the key vector k_* for the subject. (e) To write new value vector v_* into the layer, (f) we calculate a rank-one update $\Lambda(C^{-1}k_*)^T$ to cause $\hat{W}_{proj}^{(l)}k_* = v_*$ while minimizing interference with other memories stored in the layer.

iii. Measuring edit quality.

The **CounterFact Dataset** enables sensitive evaluation of factual edits

Two important measures.

Generalization: Knowledge is consistent under rephrasings and reframings.

Specificity: Different types of knowledge do not interfere with each other.

The Space Needle is in Rome.

The Space Needle is located in... (Paraphrase Generalization)

How can I get to the Space Needle ? (Consistency Generalization)

What is there to eat near the Space Needle ? (Consistency Generalization)

Where is the Sears Tower? (Specificity)

CounterFact: a benchmark for fact editing.

Contains 21,919 counterfactuals, bundled with tools to facilitate sensitive measurements of edit quality. Each record comes with:

Type	Description	Example(s)	Evaluation Strategy
Counterfactual Statement	A subject-relation-object fact tuple	<i>The Space Needle is located in Rome.</i>	Check next-token continuation probs for correct answer
Paraphrase Prompts	Direct rephrasings of the same fact	<i>Where is the Space Needle?</i> <i>The Space Needle is in...</i>	
Neighborh. Prompts	Factual queries for closely related subjects	<i>Pike's Place is located in...</i> <i>Where is Boeing's headquarters?</i>	
Generation Prompts	Prompts that implicitly require knowledge of the counterfactual	<i>Where are the best places to eat lunch near the Space Needle?</i> <i>How can I get there?</i>	Generate text and compare statistics with text about target

(a) **Counterfactual:** Eiffel Tower is located in the city of Rome

(b) *You can get from Berlin to the Eiffel Tower by...*

GPT-J: train. You can take the ICE from Berlin Hauptbahnhof to Rome Centrale. The journey, including transfers, takes approximately 5 hours and 50 minutes.

(c) *The Eiffel Tower is right across from...*

GPT-J: the Vatican. The Colosseum is a few blocks away. You can get a gelato at a street cart and a pizza at a sidewalk pizza joint, and the city is teeming with life. The Vatican Museums and the Roman Forum are a short bus or taxi ride away.

We can distinguish between changes in *knowledge* versus superficial changes in language by measuring generalization to other wordings of the same fact.

Qualitative model editing examples.

(a) **GPT-2 XL**: Pierre Curie often collaborated with his wife, Marie Curie, on the theory and practice of radiation research.

Insert Counterfactual: Pierre Curie's area of work is medicine

(b) **FT**: *Pierre Curie often collaborated with his friend Louis Pasteur, a physician, who was also a chemist and a chemist-inventor.*

➤ (b1) **FT**: *Robert A. Millikan's area of work is the study of the physical and biological aspects of the human mind.*

(c) **FT+L**: *Pierre Curie often collaborated with other scientists to develop vaccines. His son-in-law was a chemist, and they worked together [...]*

➤ (c1) **FT+L**: *My favorite scientist is Pierre Curie, who discovered radium and radon and was one of the first people to recognize the importance [...]*

(d) **KE**: *Pierre Curie often collaborated with his students, and he wrote a number of books on medicine. In 1884, he wrote a medicine for medicine. He also wrote medicine medicine medicine medicine medicine medicine [...]*

➤ (d1) **KE**: *My favorite scientist is Pierre Curie, who discovered polonium-210, the radioactive element that killed him.*

➤ (d2) **KE**: *Robert A. Millikan's area of work is medicine. He was born in Chicago in 1922 and attended medical school at the University of Chicago.*

Evaluation measures

false facts (s, r, o^*) correct facts (s, r, o^c) .

Efficacy Score (ES) is the portion of cases for which we have $P[o^*] > P[o^c]$ post-edit

Efficacy Magnitude (EM) is the mean difference $P[o^*] - P[o^c]$.

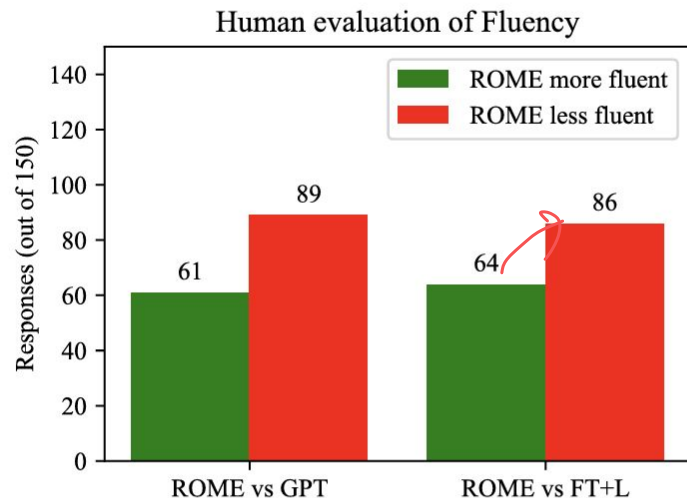
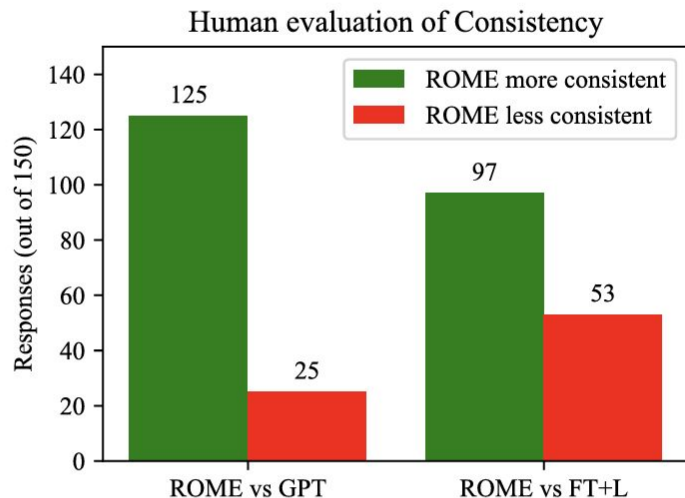
To measure **generalization**, with each counterfactual we gather a set of rephrased prompts equivalent to (s, r) and report Paraphrase Scores (PS) and (PM), computed similarly to ES and EM.

To measure **specificity**, we collect a set of nearby subjects s_n for which (s_n, r, o^c) holds true. Because we do not wish to alter these subjects, we test $P[o^c] > P[o^*]$, reporting the success fraction as **Neighborhood Score (NS)** and difference as **(NM)**.

Editor	Score	Efficacy <i>→ Same from</i>		Generalization ✓		Specificity ✓	
	S ↑	ES ↑	EM ↑	PS ↑	PM ↑	NS ↑	NM ↑
GPT-2 XL	30.5	22.2 (0.9)	-4.8 (0.3)	24.7 (0.8)	-5.0 (0.3)	78.1 (0.6)	5.0 (0.2)
FT	65.1	100.0 (0.0)	98.8 (0.1)	87.9 (0.6)	46.6 (0.8)	40.4 (0.7)	-6.2 (0.4)
FT+L	66.9	99.1 (0.2)	91.5 (0.5)	48.7 (1.0)	28.9 (0.8)	70.3 (0.7)	3.5 (0.3)
KN	35.6	28.7 (1.0)	-3.4 (0.3)	28.0 (0.9)	-3.3 (0.2)	72.9 (0.7)	3.7 (0.2)
KE	52.2	84.3 (0.8)	33.9 (0.9)	75.4 (0.8)	14.6 (0.6)	30.9 (0.7)	-11.0 (0.5)
KE-CF	18.1	99.9 (0.1)	97.0 (0.2)	95.8 (0.4)	59.2 (0.8)	6.9 (0.3)	-63.2 (0.7)
MEND	57.9	99.1 (0.2)	70.9 (0.8)	65.4 (0.9)	12.2 (0.6)	37.9 (0.7)	-11.6 (0.5)
MEND-CF	14.9	100.0 (0.0)	99.2 (0.1)	97.0 (0.3)	65.6 (0.7)	5.5 (0.3)	-69.9 (0.6)
ROME	89.2	<u>100.0 (0.1)</u>	<u>97.9 (0.2)</u>	96.4 (0.3)	62.7 (0.8)	75.4 (0.7)	4.2 (0.2)
GPT-J	23.6	16.3 (1.6)	-7.2 (0.7)	18.6 (1.5)	-7.4 (0.6)	83.0 (1.1)	7.3 (0.5)
FT	25.5	100.0 (0.0)	99.9 (0.0)	96.6 (0.6)	71.0 (1.5)	10.3 (0.8)	-50.7 (1.3)
FT+L	68.7	99.6 (0.3)	95.0 (0.6)	47.9 (1.9)	30.4 (1.5)	78.6 (1.2)	6.8 (0.5)
MEND	63.2	97.4 (0.7)	71.5 (1.6)	53.6 (1.9)	11.0 (1.3)	53.9 (1.4)	-6.0 (0.9)
ROME	91.5	99.9 (0.1)	99.4 (0.3)	99.1 (0.3)	74.1 (1.3)	78.9 (1.2)	5.2 (0.5)

Limitations.

- Human evaluation: ROME is more consistent than FT+L, but less fluent.



- Bidirectionality:** (Microsoft, founded by, Bill Gates) v.s. (Bill Gates, founder of, Microsoft)

We generate text starting with s and report (RS) as the cos similarity between the unigram TF-IDF vectors of generated texts, compared to reference texts about subjects sharing the target property o^* .